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School of Electronics Engineering (SENSE)

PROJECT – REPORT

COURSE CODE / TITLE	BECE320E Embedded C Programming	
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PROJECT TITLE	**Binnect: Edge Intelligence for Automated Waste Segregation and Monitoring**	

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OBJECTIVE:

The main objective of the Binnect project is to develop a cost-effective, real-time automated waste segregation and bin monitoring system using Edge AI and IoT. The system aims to classify waste into wet and dry categories using on-device machine learning inference on a Raspberry Pi 5, thereby eliminating the need for cloud connectivity for classification. Additionally, the system monitors bin fill levels using an ESP8266-based IoT node with ultrasonic sensors and provides a live web-based dashboard for waste management monitoring.

COMPONENTS / SOFTWARE REQUIRED:

S.No	Hardware / Software	Specification / Version
1	Raspberry Pi 5	4GB RAM, 64-bit ARM Cortex-A76
2	ESP8266 (NodeMCU)	ESP-12E module, 80MHz
3	Pi Camera Module	CSI interface, 8MP
4	HC-SR04 Ultrasonic Sensor	Range: 2cm – 400cm
5	LED Indicators (Red/Green)	5mm, GPIO-controlled
6	TensorFlow Lite	tflite-runtime 2.x
7	Python	3.9+
8	OpenCV	opencv-python 4.x
9	Picamera2 Library	Latest stable
10	RPi.GPIO	0.7.x
11	NumPy	1.23+
12	Arduino IDE / ESP8266 Framework	ESP8266WiFi, ESPAsyncWebServer

BLOCK DIAGRAM:

System Architecture – Distributed Two-Node Design:

Node 1: Edge Inference Node (Raspberry Pi 5)

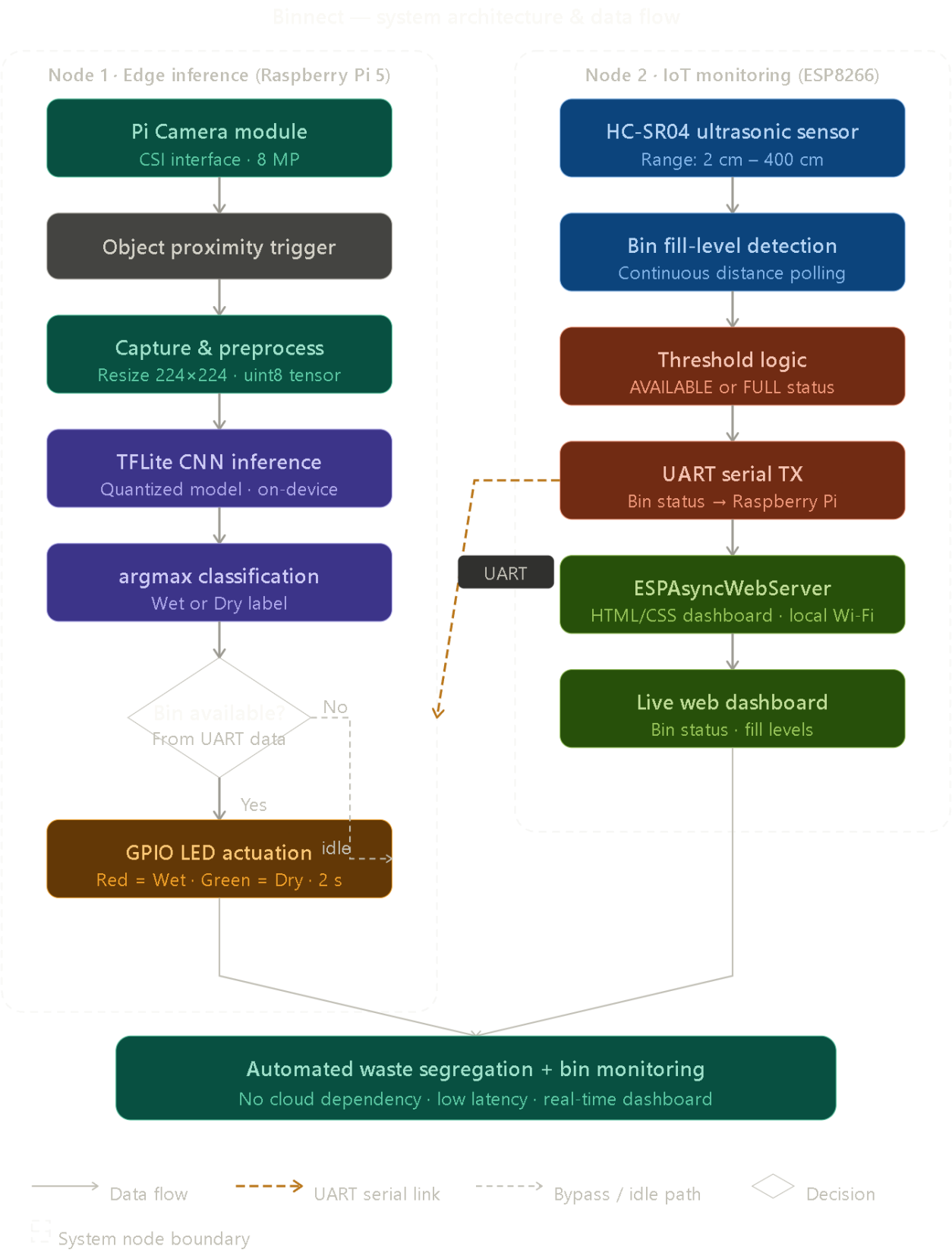
- Pi Camera → TFLite Inference Engine → Wet/Dry Classification → GPIO LED Actuator

Node 2: IoT Monitoring Node (ESP8266)

- HC-SR04 Ultrasonic Sensor → Bin Level Detection → Web Dashboard (HTML/CSS)

Inter-Device Communication: UART Serial / HTTP REST

Data Flow: Object Detected → Image Captured → Preprocessed (224x224) → TFLite Inference → argmax Classification → LED Actuation (if bin available) → Dashboard Update


Binnect — system architecture & data flow
Node 1 · Edge inference (Raspberry Pi 5)
Pi Camera module
CSI interface · 8 MP
Object proximity trigger
Capture & preprocess
Resize 224×224 · uint8 tensor
TFLite CNN inference
Quantized model · on-device
argmax classification
Wet or Dry label
Bin available?
From UART data
No
Yes
GPIO LED actuation
Red = Wet · Green = Dry · 2 s
idle
Node 2 · IoT monitoring (ESP8266)
HC-SR04 ultrasonic sensor
Range: 2 cm – 400 cm
Bin fill-level detection
Continuous distance polling
Threshold logic
AVAILABLE or FULL status
UART serial TX
Bin status → Raspberry Pi
UART
ESPAsyncWebServer
HTML/CSS dashboard · local Wi-Fi
Live web dashboard
Bin status · fill levels
Automated waste segregation + bin monitoring
No cloud dependency · low latency · real-time dashboard
Data flow
UART serial link
Bypass / idle path
Decision
System node boundary

PROJECT DESCRIPTION:

Binnect is a distributed embedded system that integrates Edge Artificial Intelligence and the Internet of Things for real-time waste classification and bin-level monitoring. The project addresses the growing challenge of manual waste segregation by automating the classification process at the point of disposal, thereby improving hygiene and efficiency.

The system is built on two primary hardware nodes working in coordination. The first is the Edge Inference Node built on a Raspberry Pi 5. When an object is placed near the bin, a proximity detection event is triggered. The Pi Camera Module captures an image, which is then preprocessed to a 224x224 resolution tensor in uint8 format. A quantized Convolutional Neural Network model, trained using Transfer Learning and deployed as a TensorFlow Lite model, performs inference entirely on-device. The argmax of the output probability vector determines whether the waste is Wet or Dry. Based on the result and the bin availability status received from the IoT node, a corresponding GPIO-connected LED is activated for 2 seconds.

The second node is the IoT Monitoring Node built on an ESP8266 (NodeMCU). It uses an HC-SR04 ultrasonic sensor to continuously measure the fill level of each bin compartment. The ESP8266 runs threshold-based logic to classify bin status as AVAILABLE or FULL and hosts a lightweight real-time web dashboard using HTML and CSS, accessible over the local Wi-Fi network. It communicates bin availability data to the Raspberry Pi via UART serial communication.

The overall data flow pipeline is: object detected via ultrasonic sensor → image captured by Pi Camera → preprocessed and fed to TFLite interpreter → class predicted using argmax → conditional LED actuation based on bin availability → dashboard refreshed with latest bin status. The entire classification pipeline operates at the edge without any cloud dependency, ensuring low latency and privacy.

CONCEPT LEARNED:

- Edge AI and on-device inference using TensorFlow Lite (TFLite) on resource-constrained hardware
- Transfer Learning for training Convolutional Neural Networks with limited data
- Model quantization (uint8) for deploying optimized models on embedded systems
- Raspberry Pi GPIO programming for LED actuation and sensor interfacing
- Camera integration using the Picamera2 library and CSI interface
- Image preprocessing pipeline: resizing, normalization, channel correction, and tensor preparation
- ESP8266 programming using the Arduino framework for IoT applications
- Ultrasonic sensor (HC-SR04) interfacing for distance and fill-level measurement
- Hosting a web dashboard on a microcontroller using ESPAsyncWebServer
- Inter-device serial communication (UART) between Raspberry Pi and ESP8266
- Distributed embedded system architecture design

- Threshold-based decision logic for real-time actuator control

IMPLEMENTATION:

The implementation was carried out in two parallel workstreams: the Edge AI inference pipeline on the Raspberry Pi 5 and the IoT monitoring firmware on the ESP8266.

Phase 1: ML Model Development

- A dataset of wet and dry waste images was collected and labelled.
- A pre-trained CNN model was fine-tuned using Transfer Learning for binary waste classification.
- The trained model was quantized to uint8 format and exported as a TFLite (.tflite) file for deployment.
- The model was validated for accuracy before deployment.

Phase 2: Raspberry Pi Edge Inference Pipeline

- The TFLite model was deployed on the Raspberry Pi 5 using the tflite-runtime library.
- The Picamera2 library was used to interface the Pi Camera and capture frames on trigger.
- Preprocessing steps including resizing to 224x224, normalization, and BGRA to BGR channel correction were implemented in Python using OpenCV and NumPy.
- GPIO pins were configured for LED actuation corresponding to wet and dry classification outputs.

Phase 3: ESP8266 IoT Firmware

- The ESP8266 firmware was developed using the Arduino IDE with the ESP8266WiFi and ESPAsyncWebServer libraries.
- HC-SR04 ultrasonic sensors were interfaced for bin fill-level detection.
- Threshold logic was programmed to toggle bin status between AVAILABLE and FULL.
- A real-time HTML/CSS web dashboard was embedded in the firmware to display bin statuses.

Phase 4: Integration and Testing

- UART serial communication was established between the Raspberry Pi and ESP8266 for bin availability data exchange.
- End-to-end testing was performed across three scenarios: object detected with bin available, object detected with bin full, and no object detected (idle state).

GitHub Repository: https://github.com/PiyushArora13/Binnect-Edge-Intelligence-for-AutomatedWaste-Segregation-and-Monitoring

CHALLENGES FACED:

- Achieving reliable real-time inference speed on the Raspberry Pi 5 within acceptable latency for a user-facing system.
- Collecting and annotating a sufficiently diverse dataset of wet and dry waste images for accurate model training.
- Handling variability in lighting conditions and camera angles which affected image preprocessing and classification accuracy.
- Debugging channel format mismatches (BGRA vs BGR) when processing frames from the Picamera2 library.
- Synchronizing communication between the Raspberry Pi and ESP8266 over UART to prevent race conditions in decision logic.
- Optimizing model quantization to uint8 without significant loss in classification accuracy.
- Managing memory constraints on the ESP8266 while hosting the web dashboard and running sensor polling simultaneously.
- Ensuring stable Wi-Fi connectivity on the ESP8266 for uninterrupted dashboard updates.

APPLICATIONS:

- Smart municipal waste management systems in cities and residential complexes for automated segregation at source.
- Industrial waste sorting in manufacturing facilities to separate recyclable and non-recyclable materials.
- Hospital and biomedical waste classification to distinguish between hazardous and nonhazardous waste.
- Educational institutions and offices implementing green campus initiatives.
- Integration with smart city infrastructure for centralized waste monitoring and route optimization for garbage collection.
- Food processing units for automated wet waste (organic) separation from dry waste (packaging).
- Extension to multi-class classification for segregating plastics, metals, glass, and organic waste.
- Remote bin monitoring in public spaces, parks, and transit hubs using IoT dashboards.

CONCLUSION:

The Binnect project successfully demonstrates a practical, low-cost, and scalable solution for automated waste segregation using Edge AI and IoT. By deploying a quantized TFLite CNN model directly on the Raspberry Pi 5, the system achieves real-time waste classification into wet and dry categories without any dependency on cloud infrastructure, making it suitable for areas with limited network connectivity.

The ESP8266-based IoT node complements the Edge AI subsystem by providing continuous bin-level monitoring and a live web dashboard for remote status visibility. The integration of both nodes through UART serial communication demonstrates a well-designed distributed embedded system architecture.

This project provided valuable hands-on experience in the intersection of Microcontrollers, Edge AI, Computer Vision, and IoT – all core domains covered under BECE320E Embedded C Programming. The modular and scalable design of Binnect allows for future enhancements such as servo-based automated lid control, MQTT-based cloud integration, mobile application interfaces, and multi-class waste classification to further improve its real-world applicability.

DEMONSTATION VIDEO:

https://drive.google.com/file/d/1j3AYxH23SDRsSLVr7zUnk9_wCZ3bVPa7/view?usp=sharing

DATASET LINK:

https://www.kaggle.com/datasets/aashidutt3/waste-segregation-image-dataset